PRODUCT REQUIREMENTS DOCUMENT

AI Customer Support Suite

Version: v2.4.7 | **Status:** APPROVED | **Date:** 2026-11-06

1. Document Information

AI Customer Support Suite - Product Requirements Document

- **Version:** v2.4.7
- **Status:** Authoritative Specification
- **Content Hash (SHA-256):** `mockhash1234567890abcdef1234567890abcdef1234567890abcdef12345678`
- **Author:** Principal Requirements Engineer
- **Domain:** Customer Support

2. Executive Summary

This document defines the authoritative engineering requirements for AI Customer Support Suite operating within Customer Support.

3. Product Overview

Detailed specifications for product_overview in Customer Support addressing operational workflows and stakeholder capabilities.

4. Problem Statement

Detailed specifications for problem_statement in Customer Support addressing operational workflows and stakeholder capabilities.

5. Product Vision and Objectives

Detailed specifications for product_vision_and_objectives in Customer Support addressing operational workflows and stakeholder capabilities.

6. Target Users and Stakeholders

Detailed specifications for target_users_and_stakeholders in Customer Support addressing operational workflows and stakeholder capabilities.

7. User Personas

- **Customer:** Inquires about product troubleshooting via self-contained web chat.
- **Customer Service Representative:** Handles complex inquiries and low-confidence escalations with 3-bullet context summary.

8. Existing Workflow and Pain Points

Detailed specifications for existing_workflow_and_pain_points in Customer Support addressing operational workflows and stakeholder capabilities.

9. Proposed Solution

Detailed specifications for proposed_solution in Customer Support addressing operational workflows and stakeholder capabilities.

10. Product Scope

- Grounded knowledge-base question and answer retrieval
- Multi-turn conversation session history management
- Low-confidence answer detection below 80.0% threshold
- Automated delivery suppression for answers < 80.0%
- Human customer service representative escalation routing with 3-bullet summary

11. Out of Scope

- External third-party ticketing platforms (Zendesk, Jira)
- External CRM synchronization (Salesforce, HubSpot)
- Telephony, voice calls, and phone hotline IVR
- Multilingual speech translation

12. Functional Requirements

ID	Title	Description	Priority	Traceable AC
FR-001	Grounded Knowledge-Base Q&A	Retrieves answers exclusively from approved knowledge-base articles	Must Have	AC-001, AC-002
FR-002	Conversation Session History	Preserves multi-turn conversation context across dialogue turns in active session memory	Must Have	AC-003, AC-004
FR-003	Low-Confidence Answer Detection	Evaluates confidence against 80.0% threshold and routes low-confidence answers to human escalation	Must Have	AC-005, AC-006

ID	Title	Description	Priority	Traceable AC
FR-004	Human Agent Escalation Routing	Transfers full transcript and structured 3-bullet context summary to human representative	Must Have	AC-007, AC-008
FR-005	Escalation Resolution Ledger	Commits agent disposition notes and session resolution timestamp to audit ledger	Should Have	AC-009

Detailed Functional Requirement Specifications

FR-001: Grounded Knowledge-Base Q&A

- **Actor:** Customer
- **System Behavior:** Retrieves approved passages and formats verified response.
- **Provenance:** `confirmed_user_input` (User Input: answers customer questions using approved knowledge-base articles)

FR-002: Conversation Session History

- **Actor:** Customer
- **System Behavior:** Preserves multi-turn dialogue context in session cache; long-term archival duration is an open decision.
- **Provenance:** `confirmed_user_input` (User Input: maintains conversation history)

FR-003: Low-Confidence Answer Detection

- **Actor:** AI Evaluation Engine
- **System Behavior:** Suppresses answers below 80.0% confidence and routes to internal escalation queue.
- **Provenance:** `confirmed_user_input` (User Input: identifies low-confidence answers below 80%)

13. Non-Functional Requirements

ID	Title	Description	Classification	Priority
NFR-001	Transaction Response Latency	System targets 95th percentile response times under 2.5s	Proposed target	Should Have
NFR-002	Session Dialogue Encryption & Security	All domain data encrypted via TLS 1.3 in transit and AES-256 at rest	Confirmed requirement	Must Have
NFR-003	High Availability & Resilient Failover	Service targets 99.5% operational uptime with automated logging	Proposed target	Should Have

14. User Stories

Detailed specifications for user_stories in Customer Support addressing operational workflows and stakeholder capabilities.

15. User Journeys

1. **Customer Query Ingestion:** Customer submits question in web chat; system retrieves matching approved KB passages.
2. **Threshold Evaluation:** Evaluates candidate answer confidence against 80.0% threshold; if >= 80.0%, delivers grounded response with article citations.
3. **Low-Confidence Escalation:** If confidence is strictly below 80.0%, system suppresses direct response, compiles 3-bullet summary, and transfers to human representative queue.
4. **Agent Resolution:** Representative reviews complete transcript with summary and resolves customer inquiry.

16. Business Rules

ID	Rule Title	Operational Statement	Provenance	Source Reference	Rationale	Confirmation Required
BR-001	Grounded Knowledge Base Answers	Answers must be derived strictly from approved articles with source citations	`confirmed_user_input`	User Input: approved knowledge-base articles	Prevents hallucinations	No (Confirmed)
BR-002	80% Confidence Escalation Threshold	Responses with confidence score below 80.0% must be suppressed and escalated	`confirmed_user_input`	User Input: identifies low-confidence answers below 80%	Protects customer trust	No (Confirmed)
BR-003	Multi-Turn Context Retention	Chat session history must be preserved across conversation turns	`confirmed_user_input`	User Input: maintains conversation history	Contextual continuity	No (Confirmed)
BR-004	Agent Handoff Summary	Escalated sessions must deliver complete transcript and 3-bullet summary to representative	`confirmed_user_input`	User Input: escalates complex queries with summary	Reduces triage latency	No (Confirmed)
BR-005	Chat Retention Policy (Open Decision)	Archival transcript retention duration beyond 24-hour cache is subject to privacy review	`unresolved_decision`	Unresolved Decision: Data Retention Policy	Statutory compliance	Yes (Review needed)

17. System Workflows

Detailed specifications for system_workflows in Customer Support addressing operational workflows and stakeholder capabilities.

18. Data Requirements

- Core entities: CustomerSession, ChatMessage, KBArticleReference, EscalationTicket, ResolutionAudit.
- Active chat session state is cached up to 24 hours; archival retention duration is documented as an Open Architectural Decision pending privacy review.

19. Integration Requirements

Detailed specifications for integration_requirements in Customer Support addressing operational workflows and stakeholder capabilities.

20. Security and Privacy Requirements

- Role-Based Access Control (RBAC) enforced across all endpoints.
- Data encrypted in transit using TLS 1.3 and at rest using AES-256.
- Statutory regulatory compliance (GDPR, FERPA, HIPAA) marked as Open Decision pending institutional legal review.

21. Acceptance Criteria

AC ID	Associated Requirement	Acceptance Criteria (Given-When-Then / Rule)
AC-001	FR-001	Given a customer submits a question, When evaluated against the knowledge base, Then answers cite approved source article references.
AC-002	FR-001	Given an answer candidate with no approved KB article, When evaluated, Then automated answer is suppressed and customer offered human escalation.
AC-003	FR-002	Given a multi-turn chat session, When customer asks a follow-up question, Then conversation context from previous turns is preserved.
AC-004	FR-002	Given an active chat session, When session continues across dialogue turns, Then transcript continuity is maintained in session memory.
AC-005	FR-003	Given an answer candidate with evaluated confidence of 79.9% (< 80.0%), When evaluated, Then automatic response is prohibited and query is escalated to a human agent with conversation summary.
AC-006	FR-003	Given an answer candidate with evaluated confidence of exactly 80.0%, When evaluated, Then automatic response is permitted and delivered to customer.
AC-007	FR-004	Given an answer candidate with evaluated confidence of 80.1%, When evaluated, Then automatic response is permitted with verified article citations.
AC-008	FR-004	Given an escalation event occurs, When transferred to a representative, Then the agent dashboard receives the full transcript and structured 3-bullet summary without prompting customer to re-enter issue details.
AC-009	FR-005	Given a human representative completes inquiry handling, When ticket is resolved, Then disposition notes and handling duration are committed to permanent audit ledger.

22. Dependencies and Constraints

Detailed specifications for dependencies_and_constraints in Customer Support addressing operational workflows and stakeholder capabilities.

23. Risks and Mitigation

Detailed specifications for risks_and_mitigation in Customer Support addressing operational workflows and stakeholder capabilities.

24. Assumptions and Open Questions

Detailed specifications for assumptions_and_open_questions in Customer Support addressing operational workflows and stakeholder capabilities.

25. Success Metrics

Detailed specifications for success_metrics in Customer Support addressing operational workflows and stakeholder capabilities.

26. Release Considerations

Detailed specifications for release_considerations in Customer Support addressing operational workflows and stakeholder capabilities.

27. Appendix and Glossary

Detailed specifications for appendix_and_glossary in Customer Support addressing operational workflows and stakeholder capabilities.